

REASONING AND LOGIC

It's safe to say that the most describable aspect of our intelligence is reasoning and logic, and thus, it is the part most easily artificially replicated. Computers are built on a foundation of logic, which is why they are able to beat us at, say Chess and Go – logically-driven games.

Even the most intelligent AI robot today, if you permit us the over simplification, still just follows lines – as our skills in coding increase the lines are becoming ever more complicated curves, but they're still directives. But who's to say humans don't work exactly the same way, right?

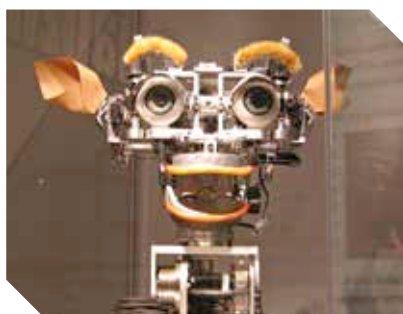
The AI that exists today is logic-based Narrow AI, and excels in domains such as Chess or Go (as we mentioned earlier), but pales in comparison to humans, for example, in the area of facial recognition. The Turing Test still holds the key for General AI – testing whether or not an AI can successfully convince humans that they are interacting with another human. But are we there yet?

ARE WE THERE YET?

The first industrial revolution was steam powered, the second was electricity and petroleum driven, the third was internet and robotics/automation caused, but a fourth is coming in the form of AI. But why not as a subset of the third revolution?

The Fourth Industrial Revolution will be due to all our advancements in nanotechnology, biotechnology, robotics, 3D printing, IoT, and of course AI. However, each industrial revolution created new job categories, will the fourth take them all back?

If we arrive at AI that is capable of mimicking human intelligence, but



Faking it: Kismet the emotive robot

can be tied into all of the advantages of robotics, you arrive at super productive individuals who never tire, never eat, need not be paid, and cost much less than the average human in the long run. The need to make a buck and keep shareholders happy should see AI replace humans in every field, right?

Even without strong AI (one that can pass the Turing Test), robotics and AI are already taking away thousands of jobs. In fact in a recent meeting, the World Economic Forum discussed this very issue. The bulk of the world's stock trading is being carried out by Narrow AI whose algorithms react to changes faster than human thought. Companies like Ross Intelligence are already using AI to enhance their paralegal practice, as software can look through a ton of past files (as long as they are digitized) faster than any human associate or intern, and makes less mistakes too. Self-driving cars are going through hot and cold phases, but we can all see the eventuality of it being one of the first human jobs to be made redundant.

Even in healthcare, an industry one would think would be the least affected by AI and robots, doctors are off-loading the tedious task of looking at a patient's genetic history to determine the most suitable drug (from hundreds available) for anti-cancer treatment. You can't even be sure that you've been chatting with a human customer care agent these days.

Creativity and ingenuity is where AI fails however. When faced with a new problem, the human mind is capable of thinking "out of the box" to find a

solution, or even to take a call on a path that incurs losses, but minimises overall damage. AI is terrible at this (so far).

Artificial Creativity can, however, be programmed by unnatural selection. Just like the evolution of life molecules, AI could evolve (a certain problem, if not itself) by making as many dissimilar random changes as possible and applying that to the problem-case to see if the solution is a better fit than the existing solution. This would be a kind of creativity captured mathematically. There is already research into AI that makes Art, thanks to Google. Neural networks, deep learning algorithms and scruffy techniques are paving the way, as evidenced by an AI developed at MIT that can 'dream' based on a given image (after having gone through millions of videos on Flickr), or the AIs at Google

Brain who developed their own encryption systems by themselves.

NOW WHAT?

Taken to its extreme, assume that AI can indeed replace humans at most professions... Let's say Apple completely replaces all humans in its company (except say, designers and technical staff) with robots, as does every other big company...

With so much unemployment, who is really going to buy an iPhone? Of course there will always be rich people, but will they be the teeming millions who buy their phones now? They sell about 80 million iPhones a quarter, and if they were reduced to 1 million buyers, and even after their costs went down, they'd still need to price the iPhone much higher to make the same profit. Weak AI is good for business and bottom lines, but strong AI that makes billions jobless is terrible for business! You can count on greed to reduce the pay that you get, but their greed will also keep you employed and seeking retail therapy.

For now, it seems like acquiring multiple skills and using creativity will keep you employed for the longest... We're just dreading the day they teach AI to write about technology... 




Guess which dog can do more tricks?